Commentary: Signatures of unconventional superconductivity near reentrant and fractional quantum anomalous Hall insulators

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Overview

This commentary is on the paper "Signatures of unconventional superconductivity near reentrant and fractional quantum anomalous Hall insulators" by F. Xu et al., first posted to ArXiv in April 2025 [1].

This paper primarily studies a dual-gated twisted homobilayer $MoTe_2$ ($tMoTe_2$) sample with a twist angle of 3.83°. The dual-gated structure allows for independent control of the carrier density, n, and the displacement field, D, through the 2-dimensional $MoTe_2$ layers. The twist between the two $MoTe_2$ layers forms a moiré pattern, introducing a superlattice potential with a moiré lattice constant of 5.27 nm. Associated with the moiré superlattice is a moiré filling factor, ν, proportional to n, denoting the filling of the moiré unit cell. The moiré superlattice along with interlayer hybridization alters the band structure, and in this case, leads to a topological flat band.

Previous works investigating $tMoTe_2$ samples at twist angles of around 3.7° to 3.9° have discovered that it hosts integer and fractional quantum anomalous Hall effects (IQAHE and FQAHE, respectively) at hole filling [2–4]. The discovery of the FQAHE in $tMoTe_2$ was the first ever observation of this phenomenon.

While the previous works include low-temperature transport studies (electron temperature $T \approx 100$ mK inside a dilution fridge) of $tMoTe_2$ with similar twist angles [4], the novelty of this work is that they found the reentrant integer quantum anomalous Hall effect (RIQAHE), an analog of the reentrant integer quantum Hall effect (RIQHE) observed in partially filled Landau levels at high magnetic fields. In addition, they discovered that superconductivity emerges from the anomalous Hall (AH) metal state between the IQAH state at $\nu = -1$ and the first FQAH state at $\nu = -2/3$. This is particularly intriguing because the superconducting state neighbors the 2/3 FQAH state, indicating a possibility of anyon superconductivity. The discovery of these novel phases can potentially be attributed to higher device quality (the authors used higher-quality crystals and an additional annealing step after stacking, along with a specific material stackup to improve contact to the $tMoTe_2$ layer), alongside high sensitivity of the observed phenomena to the twist angle.

Results

The paper begins by showing the ν-D phase diagrams of resistivity at various temperatures in Fig. 1. $\nu_h = -\nu$ is used to denote the hole filling factor, which is used throughout the work.

At $T = 2$ K (Fig. 1C), the phase diagram closely resembles that of previous studies of similar samples. Centered around $D = 0$, there are IQAH and FQAH states at $\nu_h = 1$ and $\nu_h = 2/3,\ 3/5$, respectively. These are marked by vanishing ρ_{xx} and quantized ρ_{xy} (not shown in Fig. 1 for $T = 2$ K). Applying D, topologically trivial correlated insulators emerge at $\nu_h = 1$ and various fractional fillings.

As temperature is decreased, the QAH states seem to stabilize, as seen by their regimes expanding in the phase space. Two more correlated insulators emerge at $\nu_h = 3/2$ and $\nu_h = 4/3$, which have not been previously observed, and potentially indicate higher sample quality.

More interestingly, two new types of phases appear at low temperature. First, neighboring the FQAH phases, there are pockets of vanishing ρ_{xx} and quantized ρ_{xy}, but ρ_{xy} quantized to the $\nu_h = 1$ value instead of corresponding to the actual filling. These states are the RIQAH states and highly resemble the RIQH states found in quantum Hall systems at high magnetic field. The other state that appears has vanishing ρ_{xx} and ρ_{xy}, indicating the onset of superconductivity. This state emerges from the anomalous Hall metal phase, near the border of the $\nu_h = 2/3$ FQAH state. These phases are shown more clearly in Fig. 2A–B.

The hallmark signatures of QAH states include a magnetic hysteresis loop and an n-B slope matching the Stréda formula, $\frac{\partial B}{\partial n} = -e\rho_{xy}$. These signatures are both shown in Fig. 2. While the reentrant state has the hysteresis loop between integer-quantized values of ρ_{xy}, the Stréda slope peculiarly matches the filling rather than the value of ρ_{xy}.

The observed superconducting state manifests around $D = 0$ and in the AH metal phase close to the boundary of the $\nu_h = 2/3$ FQAH state. The AH metal phase is not gapped, featuring finite ρ_{xx} and ρ_{xy}. ρ_{xy} also exhibits magnetic hysteresis, though it is not quantized. The fact that the superconductivity emerges from this polarized state is highly exotic, as it suggests that the superconductivity inherits the time-reversal symmetry breaking order of the parent AH metal state. This is supported by the magnetic hysteresis persisting through the onset of superconductivity, shown in Fig. 4. The superconducting phase also shows other classical signatures of superconductivity, including a dome of critical temperature and magnetic field (Fig. 4H), and its suppression by temperature and applied magnetic field is clearly seen in Fig. 3 and Fig. 4I–J respectively.

Looking at the temperature and magnetic field dependence also reveals information about the hierarchy and competition between the different states. The RIQAH states onset adjacent to the FQAH states at around 300 mK, while the IQAH and FQAH states are present even at 2 K. The onset of the superconducting state begins at around 1.2 K, but only reaches zero resistance at around 300 mK (Fig. 3). Additionally, at higher magnetic fields, it appears that after superconductivity is destroyed, a RIQAH state appears, overlapping with the previous superconductivity regime (Fig. 2I–J). This shows an apparent competition between the superconducting and RIQAH states.

Discussion

This paper is very exciting because it shows the interplay of various exotic phases adjacent to each other in a single sample. The presence of RIQAH states indicates the presence of zero-field charge density waves (CDWs), as the conventional explanation for the RIQHE is that some of the carriers freeze into a CDW, leaving the itinerant carrier density lower and forming an IQH state.

The superconductivity is also highly interesting because of its seemingly exotic origin, coming from a magnetic parent state. The data highly resembles the unconventional superconducting state in multilayer rhombohedral graphene, which is believed to be chiral [5]. The chiral superconductivity in rhombohedral graphene emerges from a spin- and valley-polarized metal phase, with magnetic hysteresis persisting into the superconducting phase below the critical temperature. However, the rhombohedral graphene case does not involve a moiré potential, and does not neighbor any FQAH states. Note that the FQAHE has also been observed in separate rhombohedral pentalayer graphene (R5G) samples where the graphene is aligned with hexagonal boron nitride (hBN), forming a moiré superlattice [6]. The FQAH states in aligned R5G-hBN samples even appears in the same n-D regime as the chiral superconductivity in unaligned samples, but the presence of the moiré potential fundamentally changes the phase diagram and seemingly inhibits superconductivity (though superconductivity does seem to appear at lower temperatures; see Fig. 2a of [7]).

The fact that the superconductivity shown in $tMoTe_2$ neighbors the $\nu_h = 2/3$ FQAH state is suggestive of anyon superconductivity, as the quasiparticle excitations of FQH states are anyons. However, this claim appears somewhat speculative, and more evidence and methods of probing are needed to help either substantiate or refute this claim.

This paper is also important because it suggests methods to improve sample quality for similar types of samples. The annealing step used after stacking is claimed by the authors to improve moiré uniformity and contact. However, annealing may be risky for twisted samples in general, as the high heat could lead to thermal relaxation which alters the twist angle. Presumably the $tMoTe_2$ stackup and twist angle in this case are robust against the suggested annealing parameters.

In addition to the annealing step, the samples in this paper used a contact geometry involving $TaSe_2$ in between the $tMoTe_2$ layer and the metallic Pt leads. The intermediate $TaSe_2$ between $MoTe_2$ and Pt should help circumvent the large Schottky barrier that would be present if $MoTe_2$ were to be connected to Pt directly. Furthermore, this contact geometry allows the entire heterostructure to be stacked at once and released onto a silicon chip with pre-patterned Pt leads, using the silicon as the contact gate. This workflow should improve sample quality as it avoids needing to stack the sample in multiple stages (which adds complexity and could compromise sample quality due to residue left behind between the different stages). However, in the described process, to define the Hall bar sample structure, the authors directly etched into the $MoTe_2$ layer. This can potentially degrade the quality of the $MoTe_2$ and its lifetime, as the etched edge would no longer be encapsulated.

Overall, this work showcases how high-quality $tMoTe_2$ hosts an extremely rich landscape of exotic quantum phases with complex interplay in a single sample (and even better, in a single topological flat band). There are still lots of open questions and opportunities for further studies into this system, including techniques such as optical and scanning probe measurements to corroborate the transport data. Careful characterization of the various phases, from the potential zero-field CDW crystals to the nature of the superconductivity, presents exciting directions for future research.

References

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